CSEE Statistics Section Case Study Competition

Summary

The Statistics Section Case Study Competition engages students in the analysis of large and complex ecological or evolutionary data. Teams of 2 to 5 students supported by a faculty/post-doc mentor will work together to analyse a selected data set and will present their work in poster format at the 2027 Annual Meeting of the CSEE. Posters will be judged by members of the Statistics Section, and prizes will be awarded for the best submissions.

Eligibility

The competition is open to teams of 2 to 5 combining students at any level (undergraduate, MSc, or PhD). Each team should be supported by a mentor who may be a faculty member or a post-doctoral fellow with expertise in statistical analysis or a related field. At least one member of the team must attend the conference and present their work. Limited support for travel may be available from the section.

Submission

Teams must submit an application for the case study competition through the conference website (https://event.fourwaves.com/csee-scee-2027) before the deadline for abstract submission (TBD). Enter the names of your team members and university affiliations as authors including your mentor at the end of the list. Enter “Case Study Competition” for the title and leave the abstract blank. Teams will need to follow the conferences procedures for poster submission, including submitting a digital version of your poster through the conference website before the digital upload deadline set by the Local Organizing Committee.

Case Study

Data

This year’s case study competition is supported by Dr. Danielle Ethier, Senior Scientist with Birds Canada. As part of her work monitoring bird populations in Canada, Dr. Ethier oversees the analysis of data from several of Bird Canada’s long-term citizen science projects. This year, teams will work with data from the Ontario Nocturnal Owl Survey.

Now in its 31st year, the Ontario Nocturnal Owl Survey database contains observations submitted by almost 200 volunteers annually who conduct surveys along routes across the province. Data comprise counts of owls observed along each route with associated information on environmental conditions including temperature, snow depth and cover, wind speed and cloud cover, and traffic noise levels which may impact detection rates. Teams may also choose to add further data on factors like land use or forest type from external sources. The survey primarily targets the 4 species of owl that commonly reside in Ontario (Barred Owl, Boreal Owl, Great Horned Owl, and Saw-Whet Owl).

Challenge

Evidence suggests that owl populations may be cyclic, following patterns in local resource availability (e.g., Cheveau et al. 2004). Students will be asked to develop models to understand how the populations of one or more of the four target species have varied over the past three decades. Models should describe the cycles in the population(s), determine whether they are geographically linked, identify factors they might be associated with, and/or examine whether these cycles are correlated across species. Students are encouraged to include explanatory covariates in their models.

Presentation

Posters will be displayed as part of the regular conference poster session. Your poster may be up to 76 cm wide and 91 cm tall and should include an Introduction describing the data you focused on, the questions you asked, and your hypotheses, followed by summaries of the Methods, Results, and Conclusions. Teams should also prepare a 5 to 10-minute presentation explaining their work and their conclusions to present to the judges. Teams will be judged on the quality and novelty of their analysis, the effectiveness of their poster, and the clarity of their oral presentation.

Acknowledgements

We are grateful for the support of Birds Canada and particularly thank Dr. Danielle Ethier (Senior Scientist), Dr. Kathy Jones (Volunteer Manager and Biologist), and all the volunteers who have contributed to the Ontario Owl Survey over the past 31 years.

References

Cheveau, M., Drapeau, P., Imbeau, L., & Bergeron, Y. (2004). Owl winter irruptions as an indicator of small mammal population cycles in the boreal forest of eastern North America. *Oikos*. https://doi.org/10.1111/j.0030-1299.2004.13285.x